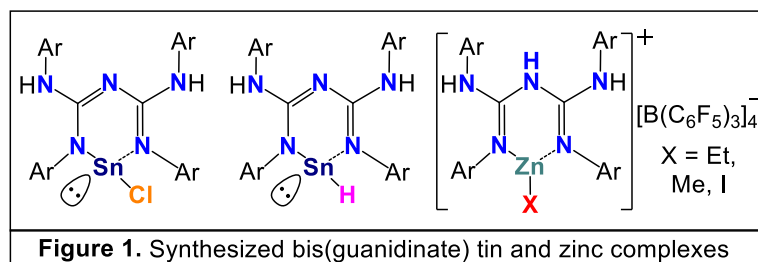
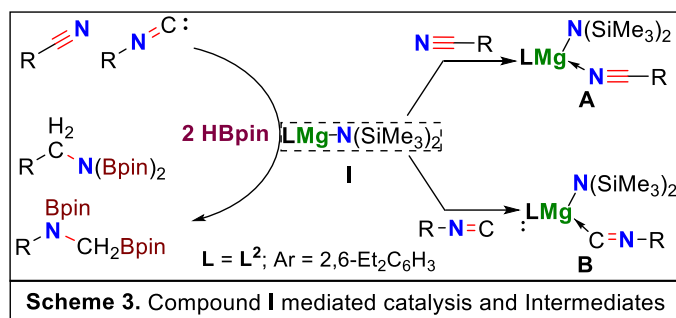
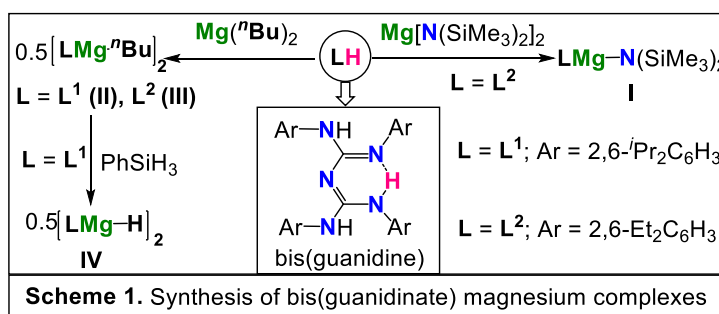


Research Highlights

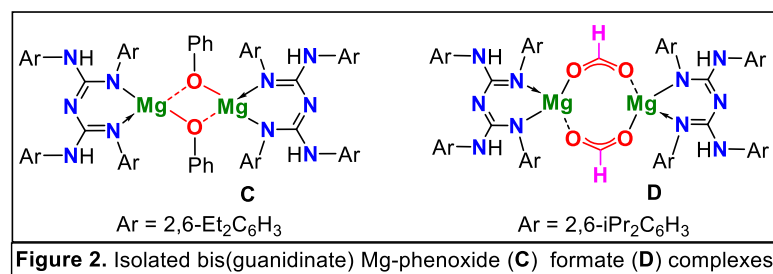


Synthesized bis(guanidinate) magnesium (II) butyl, hydride and amide complexes and employed the active hydride species for the hydroboration of cyanamides and carbodiimides. With the isolation of active species and intermediates via SCXRD established comprehensive mechanistic cycles

Synthesized Bis(guanidinate) Ligands and explored their Chemistry over reactive tin and zinc complexes. Employed them successfully for the hydroboration & cyanosilylation Reactions



We employed the magnesium amido complex for the dihydroboration of nitriles and isocyanides, and via isolation of the new nitrile and isocyanides adducts of amido magnesium complexes, established a new substrate assisted mechanism in the field. Based on the isolated intermediates and detailed kinetic study we established the mechanistic insights.



We used the reactive hydride species for the hydroboration of carbonates and carbon dioxide. With the successful isolation of magnesium aryloxide and formate species we established a detailed mechanism and shed light on the elementary steps involved

in each step of the reaction sequence.

We have also studied the reactivity pattern of the hydride species towards chalcogen transfer chemistry and established several magnesium chalcogenolate and chalcogenide species. Also, using the excellent coordination flexibility and adaptability of the bis(guanidinate) ligand system we established rare ligand stabilized magnesium bis-chalcogenolate complexes.

